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ORIGINAL PAPER

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# Strain gradient effects on the thermoelastic analysis of a functionally graded micro-rotating cylinder using generalized differential quadrature method

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**Abstract** In this paper, a strain gradient elasticity formulation for capturing the size effect in micro-scaled structures is presented to analyze the thermoelastic response of a functionally graded micro-rotating cylinder. The temperature distribution in the micro-rotating cylinder is analytically obtained by solving the steady-state, one-dimensional and axisymmetric Fourier heat conduction equation. For a functionally graded micro-rotating cylinder, except Poisson's ratio, all mechanical and thermal properties such as elastic modulus, density and thermal expansion coefficient are assumed to vary through the thickness according to a power-law distribution. The thermomechanical governing differential equation is obtained as a fourth-order ordinary differential equation in terms of mechanical displacement. The generalized differential quadrature method is used for the solution of thermal stresses, strains and displacement in the micro-rotating cylinder under internal and external pressure. At first, numerical results are presented for the micro-rotating cylinder to validate the generalized differential quadrature method. Then, the results obtained from the strain gradient elasticity are compared to the classical elasticity solution. Furthermore, numerical results illustrate the effects of non-homogeneity constant, thermal field and rotation on the distribution of Von Mises stress, Von Mises strain and radial displacement. It is perceived that the mentioned parameters have considerable effects on the distribution of stress, strain and displacement.

## 1 Introduction

Micro- and nano-systems and equipment such as micro- and nano-electromechanical systems (MEMS and NEMS) have many applications in a wide range of technologies from medical diagnostics, solar cells, to sensors [1,2]. It has been experimentally observed that the small scale effect in microstructure behavior is significant [3,4]. Since classical elasticity is not capable of predicting effects of size due to lack of a material length scale parameter [5–7], some non-classical higher-order continuum theories such as the non-local [8,9], couple stress [10–13] and strain gradient theories [5,14–17] are extended for capturing effects of size in small-scaled structures.

Many studies have been conducted over the last two decades to address the size-dependent behavior of micro- and nano-structures. Some of them are mentioned in this study. Modified couple stress theory (MCST),

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including internal material length scale parameters, was used for bending analysis of a micro-Bernoulli–Euler beam by Park and Gao [18]. They used a variational formulation based on the principle of minimum total potential energy to analyze a cantilever beam, and it was concluded that the bending rigidity predicted by the new model is larger than in the classic beam model. They also found that the difference between the deflections predicted by two models is considerable for thin beams. Ma et al. [19] expanded a microstructure Timoshenko beam model based on MSCT and Hamilton's principle. They solved static bending and free vibration of a simply supported Timoshenko beam. It was demonstrated that the deflections and rotation predicted by the new model are smaller than using the classical Timoshenko beam model in static bending problems. In addition, they found that the natural frequency predicted by the new model is higher than the classic model in free vibration problems. A size-dependent Bernoulli–Euler micro-beam model made of functionally graded materials (FGMs) was presented based on strain gradient theory (SGT) by Kahrobaiyan et al. [20]. They obtained the governing equations and both classical and non-classical boundary conditions by employing a variational method. In their work, the material properties was assumed to vary according to a power-law to study static and dynamic analysis of a simply supported functionally graded (FG) micro-beam. Also, an analytical solution for the static and dynamic analysis of a micro-cantilever Bernoulli–Euler beam was studied on the basis of strain gradient elasticity theory by Kong et al. [21]. They found that the beam deflection diminishes and the natural frequency increases when the beam thickness becomes comparable to the material length scale parameter. Using modified strain gradient theory, Hosseini and Bahaadini [22] studied size-dependent stability analysis of cantilever micro-pipes conveying fluid. In their work, governing equations and associated boundary conditions were derived by employing Hamilton's principle. Reddy [23] reported nonlinear theories of a Timoshenko Bernoulli–Euler FG beam using the principle of virtual displacements based on MCST, and the Von Karman geometric nonlinearity. He studied the effect of the material length scale parameter on static bending, vibration and beam buckling. Asghari et al. [24] investigated a size-dependent nonlinear Timoshenko micro-beam model based on SGT. They derived equations of motion and boundary conditions for the mechanical analysis of Timoshenko micro-beams with large deflections by a variational approach. Also, several studies have been conducted to investigate the size and surface effects on the vibration and stability analysis of nanotubes conveying fluid subjected to different loading and boundary conditions by Hosseini and his associates [25–28].

Vibration analysis of a micro-plate subjected to a moving load based on the MSCT in conjunction with Kirchhoff–Love plate theory was developed by Simsek et al. [29]. The equations of motion of the problem were derived using Lagrange equations and solved by using the implicit time integration Newmark-$\beta$ method. Also, Ke et al. [30] investigated a Mindlin micro-plate based on MCST. They derived the higher-order equations of motion and boundary conditions by employing Hamilton's principle and also obtained the natural frequencies of the micro-plate using the p-version Ritz method. Moreover, they studied effects of the material length scale parameter and observed that the size effect is significant when the micro-plate thickness approaches the material length scale parameter. Biaxial buckling and free vibration analysis for various numbers of connected micro- and nano-plates which were subjected to diverse loading conditions have been studied in Refs. [31–34]. In another work, Mohammadi and Fooladi Mahani [35] investigated a size-dependent Kirchhoff plate model for determining the critical buckling loads of rectangular micro-plates using modified strain gradient theory with three length scale parameters. In their work, the nonlinear form of the governing equations and boundary conditions was derived by employing a variational approach and the principle of minimum total potential energy.

Danesh and Asghari [36] analyzed the mechanical behavior of micro-rotating disks based on strain gradient elasticity. They derived the governing equations and boundary conditions using a variational method and solved the equations analytically. The variational formulation of the simplified strain gradient theory for thick-walled cylinders under pressure was presented by Gao and Park [37]. Sadeghi et al. [38] have presented strain gradient elasticity solution for a micro-cylinder made of FGMs to develop the studies by Gao and Park [37]. They considered the material properties vary through the thickness according to a power-law function and studied effects of non-homogeneity constant and characteristic length parameter on stresses and displacement distributions.

Thermoelastic analysis of FG cylinders is an interesting topic for researchers. Tokovvy and Ma [39] studied two-dimensional analysis of non-axisymmetric elasticity and thermoelasticity for a heterogeneous hollow cylinder. Their method was based on direct integration proposed by Vihak. They represented the solution of equations in Fourier series form and derived integral Volterra-type equation to determine Fourier series coefficients. Shengand Wang [40] investigated the nonlinear response of FG cylindrical shells subjected to mechanical and thermal loading by employing Von Karman nonlinear theory. They solved the nonlinear equation of motion using the fourth-order Runge–Kutta numerical method and reported influence of temperature change and non-homogeneity constant on nonlinear dynamic response of FG cylindrical shells. Peng and Li [41] presented thermoelastic analysis of FG pressure vessels. They studied a new method for analysis of thermal stresses in FG hollow cylinders. They obtained distributions of thermal stresses and radial displacement through numerical solution. Lutz and Zimmerman [42] analyzed thermal stresses and effective thermal expansion coefficient for FG spheres. They presented an exact solution for uniform heat transfer problem in spheres of which mechanical and thermal properties vary linearly as a function of the radius. In their work, an exact solution for stresses and displacement using the Frobenius series method was obtained. Also, an analysis of the thermomechanical behavior of hollow circular cylinders of functionally graded material was reported by Liew and et al. [43]. Dai and Wang [44] studied an analytical solution for magneto-thermo-electro-elastic problems of a piezoelectric hollow cylinder placed in an axial magnetic field subjected to arbitrary thermal shock, mechanical load and transient electric excitation. Also, Hosseini and Dini [45] reported an exact analysis on the magneto-thermoelastic behavior of rotating functionally graded cylinders in which the effects of rotation, thermal, magnetic field and non-homogeneity constant were studied on displacement, Von Mises equivalent stress and Von Mises equivalent strain fields.

In numerical solution of the differential equations, the differential quadrature method (DQ) is a computational solving technique which was first proposed by Bellman et al. [46]. Two-dimensional solution for free vibration analysis of parabolic shells was reported by Tornabene and Viola [47] using the generalized differential quadrature (GDQ) method. Wu and Liu [48] presented the generalized differential quadrature rule (GDQR) for solving initial-value differential equations. They used the Hermite interpolation function as trial functions to obtain weighting coefficients. Their results describe that the GDQR method has high precision. Chen et al. [49] studied a new method based on DQ method for free vibration of laminated beams. Bert and Malik [50] analyzed free vibration of thin cylindrical shells with different boundary conditions using DQ method.

To the best of the authors' knowledge in the available literature, there has been no attempt toward investigating the thermomechanical elastic response of a rotating FG micro-cylinder with considering the size effects. Understanding the elastic response of micro-cylinder could be a key point for the application in microelectronic and micro-mechanical devices such as MEMS instruments [51], micro-gyroscopes [52], micro-motors [53], micro-milling tools and micro-grinding wheels [54]. Therefore, in this study, classical strain gradient elasticity is employed for the analysis of stress, strain and displacement fields in a micro-rotating cylinder made of FGMs under thermal and mechanical loading. The mechanical and thermal properties of the micro-rotating cylinder are considered to vary along the thickness according to a power-law function. The temperature distribution is obtained in cylindrical coordinates by solving the one-dimensional Fourier heat transfer equation. The generalized differential quadrature method is employed to solve the fourth-order ordinary differential equation. Numerical results clearly show the effects of temperature field and characteristic length parameter on the distribution of stress, strain and displacement.

## 2 Classical strain gradient elasticity

In the non-classical continuum theory, Mindlin first proposed strain gradient elasticity which is the most complex form of higher-order continuum theories [55]. By modifying the theory of Mindlin, Lam et al. [5] mentioned a simple form of strain gradient theory for capturing the size effect in microstructures. The main difference between strain gradient elasticity and classical elasticity is in the strain energy density $U_0$ which, in the former, is a function of strain and second-order deformation gradient [5],

$$U_0 = U_0\left(\varepsilon_{ij}, \eta_{ijk}\right), \tag{1}$$

where $\varepsilon_{ij} = \frac{1}{2}\left(u_{i,j} + u_{j,i}\right)$ and $\eta_{ijk} = u_{k,ij}$ are strain tensor and the second-order deformation gradient tensor, respectively. $u_i$ is the displacement vector and $u_{i,j}$ is the derivative of $u_i$ with respect to $x_j$. The strain tensor has 6 independent symmetric components, and the second-order deformation gradient tensor has 18 independent components which are symmetric in the first two indices. Therefore, the simplified elastic strain energy density function for FG materials can be expressed as [37,38]:

$$U_0 = U_0\left(\varepsilon_{ij}, \varepsilon_{ij,k}\right) = \frac{1}{2}\lambda\left(x, y\right)\varepsilon_{ii}\varepsilon_{jj} + \mu\left(x, y\right)\varepsilon_{ij}\varepsilon_{ij} + l^2\left[\frac{1}{2}\lambda\left(x, y\right)\varepsilon_{ii,k}\varepsilon_{jj,k} + \mu\left(x, y\right)\varepsilon_{ij,k}\varepsilon_{ij,k}\right], \tag{2}$$

in which $\lambda(x, y)$ and $\mu(x, y)$ are the Lamé parameters as a function of two Cartesian coordinates $(x, y)$, and $l$ is the characteristic length parameter. Considering the thermal field in a micro-rotating cylinder, the components of the *Cauchy* strain tensor $\tau_{ij}$, the double stress tensor $\mu_{ijk}$, the strain gradient tensor $\kappa_{ijk}$ and total stress tensor $\sigma_{ij}$ are, respectively, defined as follows [37,38]:

$$\tau_{ij} = \tau_{ji} = \frac{\partial U_0}{\partial \varepsilon_{ij}} - (3\lambda + 2\mu)\,\alpha \Delta T, \tag{3}$$

$$\mu_{ijk} = \mu_{jik} = \frac{\partial U_0}{\partial \kappa_{ijk}}, \tag{4}$$

$$\kappa_{ijk} = \varepsilon_{ij,k} = \frac{1}{2}\left(u_{i,jk} + u_{j,ik}\right), \tag{5}$$

$$\sigma_{ij} = \tau_{ij} - \mu_{ijk,k}, \tag{6}$$

where $\alpha$ is the thermal expansion coefficient, $\Delta T = T(r) - T^*$ and $T^* = 25\,°\mathrm{C}$ is the reference temperature. Also, the equilibrium equation in strain gradient elasticity can be written as:

$$\sigma_{ij,j} + f_i = 0. \tag{7}$$

## 3 Formulation of basic equations

A functionally graded micro-cylinder is considered with inner radius $r_i$ and outer radius $r_o$. The micro-cylinder is rotating at a constant angular velocity $\omega$ about $z$-axis of the cylindrical coordinate system $(r, \theta, z)$. The FG micro-rotating cylinder is exposed to internal and external pressure $P_i$ and $P_\mathrm{o}$, respectively. The inner and outer surfaces of the FG micro-rotating cylinder are subjected to temperature changes $T_i$ and $T_o$, respectively. For the FG micro-rotating cylinder, the elastic strain energy density as a function of strain components can be written in the cylindrical coordinate system as:

$$\begin{aligned} U_0 = {} & \frac{1}{2}\lambda(r)\left[\varepsilon_{rr} + \varepsilon_{\theta\theta} + \varepsilon_{zz}\right]^2 + \mu(r)\left[\varepsilon_{rr}^2 + \varepsilon_{\theta\theta}^2 + \varepsilon_{zz}^2\right] + \frac{1}{2}\lambda(r) l^2 \left[\frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{rr} + \frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{\theta\theta} + \frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{zz}\right]^2 \\ & + \mu(r) l^2 \left[\left(\frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{rr}\right)^2 + \left(\frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{\theta\theta}\right)^2 + \left(\frac{\mathrm{d}}{\mathrm{d}r}\varepsilon_{zz}\right)^2\right] \end{aligned} \tag{8}$$

Also, hoop and radial strains can be defined in terms of displacement in the cylindrical coordinate system as:

$$\varepsilon_{rr} = \frac{\partial u}{\partial r}, \qquad \varepsilon_{\theta\theta} = \frac{u}{r}, \qquad \varepsilon_{zz} = 0. \tag{9}$$

In this paper, mechanical and thermal properties of the micro-rotating cylinder vary as a function of radius. There are different models for expressing variations of properties in radial direction: In this study, the power-law is used. Therefore, the Lamé parameters, thermal and mechanical properties are defined as follows:

$$\lambda(r) = \lambda_0 \left(\frac{r}{r_i}\right)^n, \mu(r) = \mu_0 \left(\frac{r}{r_i}\right)^n, \alpha(r) = \alpha_0 \left(\frac{r}{r_i}\right)^n, \rho(r) = \rho_0 \left(\frac{r}{r_i}\right)^n, k(r) = k_0 \left(\frac{r}{r_i}\right)^n \tag{10}$$

$\lambda_0$ and $\mu_0$ are material constants, $\alpha_0$ is thermal expansion coefficient, $\rho_0$ is density, and $k_0$ is thermal conduction coefficient, all of them, at the inner surface $(r = r_i)$ of micro-rotating cylinder and $n$ is non-homogeneity constant of FG material. Utilizing Eqs. (3)–(6) and (8)–(10), the hoop and radial stresses can be written in terms of radial displacement as follows:

$$\begin{aligned} \sigma_{rr} = {} & \lambda_0 \left(\frac{r}{r_i}\right)^n \left[\frac{\partial u}{\partial r} + \frac{u}{r}\right] + 2\mu_0 \left(\frac{r}{r_i}\right)^n \frac{\partial u}{\partial r} - \frac{\partial}{\partial r}\left[l^2 \lambda_0 \left(\frac{r}{r_i}\right)^n \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} - \frac{u}{r^2}\right)\right. \\ & \left. + 2 l^2 \mu_0 \left(\frac{r}{r_i}\right)^n \frac{\partial^2 u}{\partial r^2}\right] - (3\lambda_0 + 2\mu_0)\left(\frac{r}{r_i}\right)^{2n} \alpha_0 \Delta T \end{aligned} \tag{11a}$$

$$\sigma_{\theta\theta} = \lambda_0 \left(\frac{r}{r_i}\right)^n \left[\frac{\partial u}{\partial r} + \frac{u}{r}\right] + 2\mu_0 \left(\frac{r}{r_i}\right)^n \frac{u}{r} - \frac{\partial}{\partial r}\left[l^2 \lambda_0 \left(\frac{r}{r_i}\right)^n \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} - \frac{u}{r^2}\right)\right.$$
$$\left. + 2l^2 \mu_0 \left(\frac{r}{r_i}\right)^n \left(\frac{1}{r}\frac{\partial u}{\partial r} - \frac{u}{r^2}\right)\right] - (3\lambda_0 + 2\mu_0)\,\alpha_0 \left(\frac{r}{r_i}\right)^{2n} \Delta T \tag{11b}$$

According to Eq. (7), mechanical equilibrium equation in cylindrical coordinate is simplified as follows:

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \rho(r)\omega^2 r = 0. \tag{12}$$

## 4 Fourier heat transfer problem

To obtain stresses under thermal field effect in the FG micro-rotating cylinder, it is evident that the temperature distribution should also be described in the cylindrical coordinate system. The steady-state Fourier heat transfer equation without internal heat generation is defined as follows:

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left[r k(r) \frac{\mathrm{d}T}{\mathrm{d}r}\right] = 0. \tag{13}$$

Therefore, the temperature distribution is obtained by solving the differential equation (13) as follows:

$$T(r) = A + B r^{(-n)}. \tag{14}$$

A and B are the thermal constants and are calculated from the thermal boundary conditions (15):

$$T(r)|_{r=r_i} = T_i, \tag{15a}$$
$$T(r)|_{r=r_0} = T_o. \tag{15b}$$

By combining Eqs. (11), (12) and (14), the governing equation of the FG micro-rotating cylinder is obtained in terms of displacement based on strain gradient elasticity as:

$$F_4 \frac{\mathrm{d}^4 u}{\mathrm{d}r^4} + F_3 \frac{\mathrm{d}^3 u}{\mathrm{d}r^3} + F_2 \frac{\mathrm{d}^2 u}{\mathrm{d}r^2} + F_1 \frac{\mathrm{d}u}{\mathrm{d}r} + F_0 u = -\rho_0 \omega^2 r + \frac{(3\lambda_0 + 2\mu_0)\,\alpha_0}{r_i^n}\left[2Anr^{n-1} + nBr^{n-1}\right], \tag{16a}$$

where $F_i$ $(i = 0, 1, 2, 3, 4)$ are defined as

$$\begin{aligned}
F_0 &= [\mu_0 (4 - 2n) + \lambda_0 (n-2)(n-3)]\, l^2 r^{-4} + [\lambda_0 (n-1) - 2\mu_0]\, r^{-2},\\
F_1 &= (\lambda_0 + 2\mu_0)(n+1)\, r^{-1} + [-\lambda_0 (n-2)(n-3) + \mu_0 (2n-4)]\, l^2 r^{-3},\\
F_2 &= (\lambda_0 + 2\mu_0) + \left[2\mu_0 \left(1 - n^2\right) - \lambda_0 \left(n^2 + n - 3\right)\right] l^2 r^{-2},\\
F_3 &= -(\lambda_0 + 2\mu_0)(1 + 2n)\, l^2 r^{-1},\\
F_4 &= -(\lambda_0 + 2\mu_0)\, l^2.
\end{aligned} \tag{16b}$$

It should be pointed out that Eq. (16a) based on the simplified strain gradient theory is the fourth-order ordinary differential equation for a micro-rotating cylinder. The governing equation can be reduced to the second-order ordinary differential equation in classical elasticity by letting $l = 0$.

It is assumed that the axisymmetric FG micro-rotating cylinder is exposed to internal pressure $P_i$ and external pressure $P_o$. Therefore, the mechanical boundary conditions can be expressed as [38]:

$$\begin{aligned}
&\left\{\tau_{rr} - l^2\left[\tfrac{\mathrm{d}^2\tau_{rr}}{\mathrm{d}r^2} + \tfrac{1}{r}\left(\tfrac{\mathrm{d}\tau_{rr}}{\mathrm{d}r} - \tfrac{\mathrm{d}\tau_{\theta\theta}}{\mathrm{d}r}\right) - \tfrac{2}{r^2}\left(\tau_{rr} - \tau_{\theta\theta}\right)\right]\right\}_{r=r_i} = -P_i,\\
&\left\{\tau_{rr} - l^2\left[\tfrac{\mathrm{d}^2\tau_{rr}}{\mathrm{d}r^2} + \tfrac{1}{r}\left(\tfrac{\mathrm{d}\tau_{rr}}{\mathrm{d}r} - \tfrac{\mathrm{d}\tau_{\theta\theta}}{\mathrm{d}r}\right) - \tfrac{2}{r^2}\left(\tau_{rr} - \tau_{\theta\theta}\right)\right]\right\}_{r=r_0} = -P_0,\\
&l^2 \left.\tfrac{\mathrm{d}\tau_{rr}}{\mathrm{d}r}\right|_{r=r_i} = 0,\\
&l^2 \left.\tfrac{\mathrm{d}\tau_{rr}}{\mathrm{d}r}\right|_{r=r_o} = 0.
\end{aligned} \tag{17}$$

It can be found that the four boundary conditions of Eq. (17) are reduced to two boundary conditions in classical elasticity by eliminating the strain gradient effect ($l = 0$).

## 5 Differential quadrature method

The differential quadrature method has recently emerged as a powerful numerical discretization tool. One of the advantage of the DQ method is that the solution is converging and accurate enough with a considerably small number of discretization points, which requires less CPU time and computational effort [56]. The other advantage of the method in comparison with solution methods like variational approach, finite element methods and finite difference methods is its inherent simplicity in formulation that can be easily adopted to various boundary conditions. So, this approach has been employed by many researchers for analyzing various problems in nano- and micro-scale [57–61]. Here, the generalized differential quadrature method is applied to solve the differential equation (16) and boundary conditions (17). The nature of the GDQ method is that the partial derivatives of a function with respect to a space variable at a given point are approximated by a weighted linear summation of the function values at all discrete points in the domain. In this method, any desirable grid distribution can be selected without limitation. As a result, the $mth$-order derivative of function $f(x)$ at a discrete point $x_k$ can be approximated by the weighted linear sum of the function values as [56]

$$\left.\frac{\mathrm{d}^m f}{\mathrm{d}x^m}\right|_{x=x_k} = \sum_{j=1}^{N} A_{kj}^{(m)} f\left(x_j\right); \quad (k = 1, 2, \ldots, N), \tag{18}$$

where $A_{kj}^{(m)}$ are the weighting coefficients of the $mth$-order derivative and $N$ is the number of grid points. For the first derivative, the weighting coefficients can be obtained by the following relation [56]:

$$A_{kj}^{(1)} = \frac{L(x_k)}{\left(x_k - x_j\right) L\left(x_j\right)}; \quad (k, j = 1, 2, \ldots, N; \quad k \neq j), \tag{19}$$

in which $L(x_k)$ is defined as follows:

$$L(x_k) = \prod_{j=1, j\neq k}^{N} \left(x_k - x_j\right). \tag{20}$$

For the second- and higher-order derivatives, the weighting coefficients are evaluated by the matrix product of the weighting coefficients of the first- and lower-order ones which is one other important advantage of DQM that makes it a simple and powerful numerical method [56]. These weighting coefficients can be obtained as:

$$A_{kj}^{(m)} = m\left(A_{kk}^{(m-1)} A_{kj}^{(1)} - \frac{A_{kj}^{(m-1)}}{\left(x_k - x_j\right)}\right); \quad (k, j = 1, 2, \ldots, N;\ k \neq j;\ m = 2, 3, \ldots, N-1), \tag{21}$$

$$A_{kk}^{(m)} = -\sum_{j=1, j\neq k}^{N} A_{kj}^{(m)}; \quad (k, j = 1, 2, \ldots, N; \quad m = 2, 3, \ldots, N-1). \tag{22}$$

An effective method for selecting the grid points is achieved by using the following equation:

$$r_k = a + \frac{1}{2}\left[1 - \cos\left(\frac{(k-1)\pi}{N-1}\right)\right](b-a); \quad (k = 1, 2, \ldots, N), \tag{23}$$

where $a$ and $b$ are the first and last of the intervals. A set of algebraic equations is constructed by applying the GDQ method to the differential equation and boundary conditions. Therefore, Eq. (16) is approximated by employing the GDQ method as follows:

$$-(\lambda_0 + 2\mu_0) l^2 \sum_{j=1}^{N} A_{kj}^{(4)} U_j - (\lambda_0 + 2\mu_0)(1+2n) l^2 \sum_{j=1}^{N} \frac{1}{r_k} A_{kj}^{(3)} U_j + (\lambda_0 + 2\mu_0) \sum_{j=1}^{N} A_{kj}^{(2)} U_j$$
$$+ \left[2\mu_0\left(1-n^2\right) - \lambda_0\left(n^2+n-3\right)\right] l^2 \sum_{j=1}^{N} \frac{1}{r_k^2} A_{kj}^{(2)} U_j + (\lambda_0 + 2\mu_0)(n+1) \sum_{j=1}^{N} \frac{1}{r_k} A_{kj}^{(1)} U_j$$

$$+\left[\lambda_0\,(2-n)\,(3-n)+\mu_0\,(2n-4)\right]l^2\sum_{j=1}^{N}\frac{1}{r_k^3}A_{kj}^{(1)}\,U_j$$
$$=\left[\left(\mu_0\,(4-2n)+\lambda_0\,(n-2)\,(n-3)\right)\frac{l^2}{r_k^4}+\left(\lambda_0\,(n-1)-2\mu_0\right)\frac{1}{r_k^2}\right]$$
$$-\rho_0\omega^2 r_k+(3\lambda_0+2\mu_0)\,\alpha_0\left(2Anr_k^{n-1}+\frac{nB}{r_k}\right);\quad (k=3,4,\ldots,N-2)\,. \tag{24}$$

The following equations are obtained by applying GDQ method to the boundary conditions (17):

$$(\lambda_0+2\mu_0)r_1^2\sum_{j=1}^{N}A_{1j}^{(2)}\,U_j+\left[\lambda_0\,(n+1)+2\mu_0 n\right]r_1\sum_{j=1}^{N}A_{1j}^{(1)}\,U_j+\lambda_0\,(n-1)\,U_1$$
$$-(3\lambda_0+2\mu_0)\,\alpha_0\left(2Anr_1^{n+1}+nBr_1\right)=0;\quad (k=1)\,, \tag{25}$$
$$\frac{1}{r_i^n}\left\{\left[(\lambda_0+2\mu_0)r_2^n-l^2r_2^{n-2}\lambda_0\,(n-1)\,(n+2)+2\mu_0 l^2 r_2^{n-2}\left(n^2-2n+3\right)\right]\sum_{j=1}^{N}A_{2j}^{(1)}\,U_j\right.$$

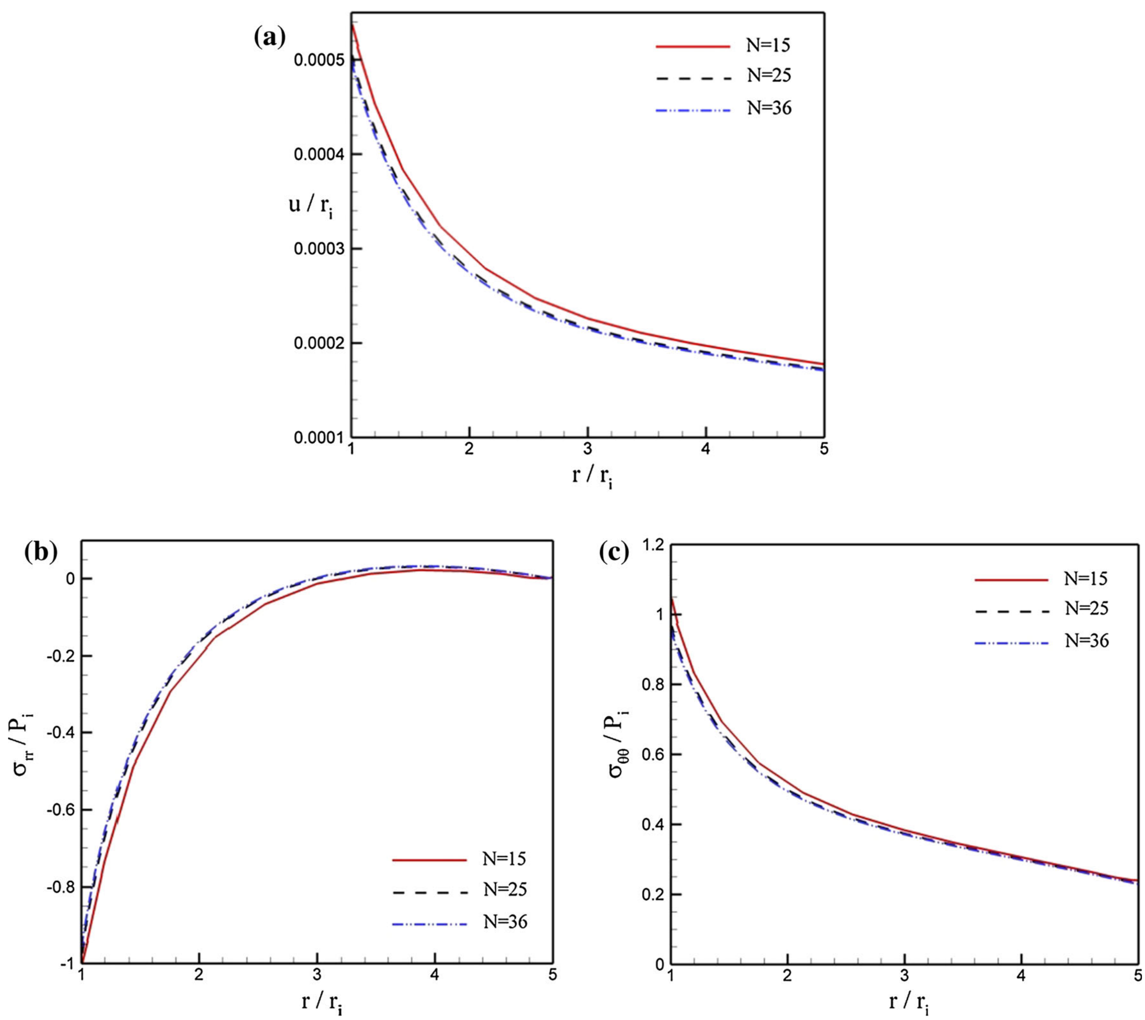


**Fig. 1** Convergence of grid point ($T_i = T_o = 0$, $P_i = 10\,\text{MPa}$, $P_o = 0$, $\omega = 4\,\text{rad}/s$, $n = 0.5$, $l = 0.1\,\mu\text{m}$). **a** The radial displacement; **b** the radial stress; **c** the hoop stress

$$- (\lambda_0 + 2\mu_0)\, l^2 r_2^n \sum_{j=1}^{N} A_{2j}^{(3)}\, U_j - (\lambda_0 + 2\mu_0)\,(2n+1)\, l^2 r_2^{n-1} \sum_{j=1}^{N} A_{2j}^{(2)}\, U_j$$

$$\left. + \left[\lambda_0 r_2^{n-1} - (n-1)(n-2)\, l^2 \lambda_0 r_2^{n-3} + 2l^2 \mu_0 r_2^{n-3}(n-3)\right] U_2 \right\} = -P_i; \quad (k=2)\,, \tag{26}$$

$$\frac{1}{r_i^n}\left\{\left[(\lambda_0 + 2\mu_0) r_{N-1}^n - l^2 r_{N-1}^{n-2} \lambda_0 (n-1)(n+2) + 2\mu_0 l^2 r_{N-1}^{n-2}\left(n^2 - 2n + 3\right)\right] \sum_{j=1}^{N} A_{N-1j}^{(1)}\, U_j\right.$$

$$- (\lambda_0 + 2\mu_0)\, l^2 r_{N-1}^n \sum_{j=1}^{N} A_{N-1j}^{(3)}\, U_j - (\lambda_0 + 2\mu_0)\,(2n+1)\, l^2 r_{N-1}^{n-1} \sum_{j=1}^{N} A_{N-1j}^{(2)}\, U_j$$

$$\left. + \left[\lambda_0 r_{N-1}^{n-1} - (n-1)(n-2)\, l^2 \lambda_0 r_{N-1}^{n-3} + 2l^2 \mu_0 r_{N-1}^{n-3}(n-3)\right] U_{N-1} \right\} = -P_{\mathrm{o}}; \quad (k=N-1)\,, \tag{27}$$

$$(\lambda_0 + 2\mu_0) r_N^2 \sum_{j=1}^{N} A_{Nj}^{(2)}\, U_j + [\lambda_0 (n+1) + 2\mu_0 n]\, r_N \sum_{j=1}^{N} A_{Nj}^{(1)}\, U_j + \lambda_0 (n-1)\, U_N$$

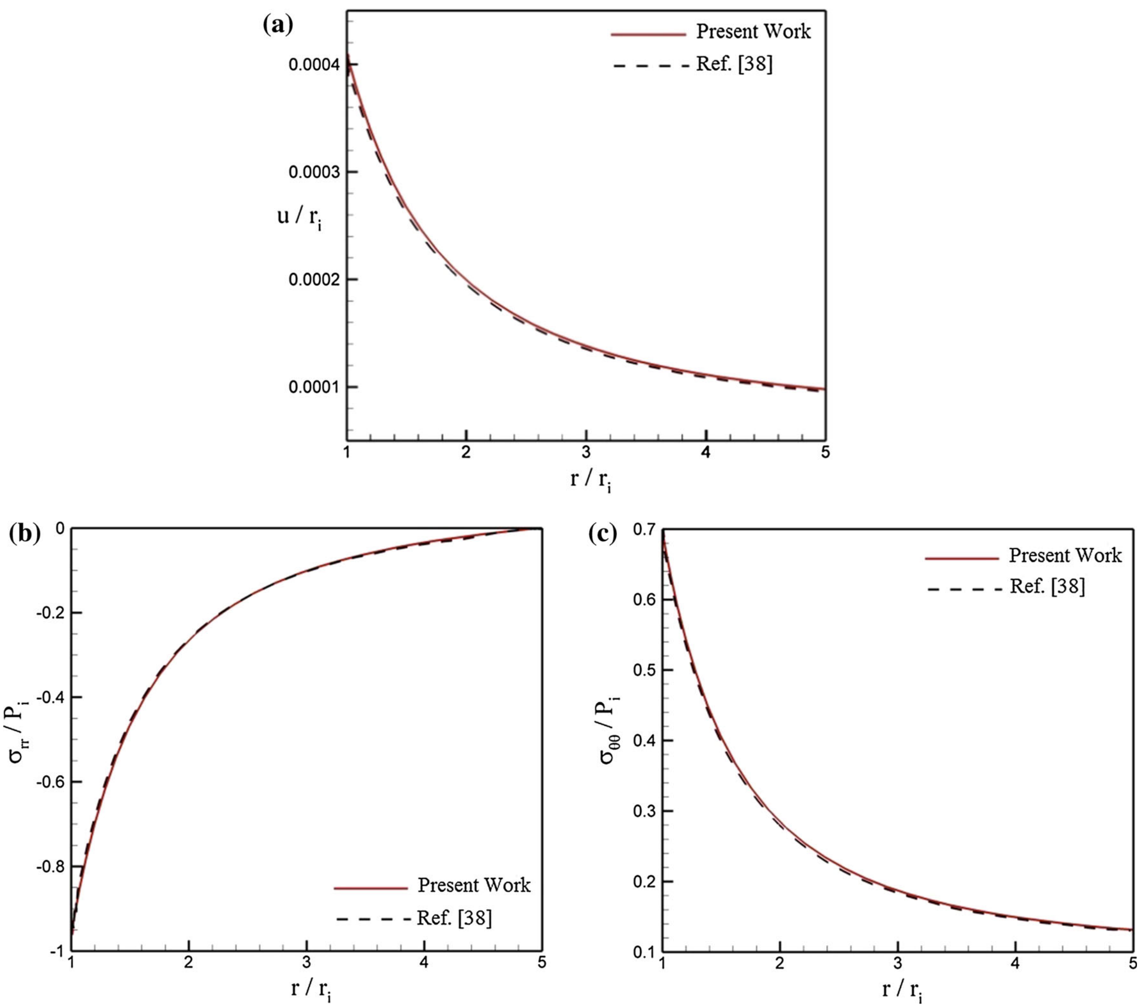


**Fig. 2** Comparison between present work and reference [38] ($T_i = T_o = 0$, $P_i = 10\,\mathrm{MPa}$, $P_{\mathrm{o}} = 0$, $\omega = 0$, $n = 0.5$, $l = 0.1\,\mu\mathrm{m}$). **a** The radial displacement; **b** the radial stress; **c** the hoop stress

$$-(3\lambda_0+2\mu_0)\,\alpha_0\left(2Anr_N^{n+1}+nBr_N\right)=0;\quad (k=N)\,. \tag{28}$$

By applying the GDQ method to radial and hoop stresses, we get:

$$\begin{aligned}\sigma_{rr}=\frac{1}{r_n^n}\Bigg\{&\lambda_0 r_k^n\left[\sum_{j=1}^{N}A_{kj}^{(1)}U_j+\frac{U_k}{r_k}\right]+2\mu_0 r_k^n\sum_{j=1}^{N}A_{kj}^{(1)}\;U_j\\&-l^2\lambda_0 n r_k^{n-1}\left[\sum_{j=1}^{N}A_{kj}^{(2)}U_j+\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(1)}U_j-\frac{U_k}{r_k^2}\right]\\&-l^2\lambda_0 r_k^n\left[\sum_{j=1}^{N}A_{kj}^{(3)}U_j+\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(2)}U_j-\sum_{j=1}^{N}\frac{2}{r_k^2}A_{kj}^{(1)}U_j+\frac{2U_k}{r_k^3}\right]\\&-2l^2\mu_0 n r_k^{n-1}\sum_{j=1}^{N}A_{kj}^{(2)}U_j-2l^2\mu_0 r_k^n\sum_{j=1}^{N}A_{kj}^{(3)}U_j\Bigg\}-\frac{(3\lambda_0+2\mu_0)\,\alpha_0 r_k^{2n}\left(A+Br_k^{-n}\right)}{r_i^{2n}},\end{aligned} \tag{29}$$

$$\begin{aligned}\sigma_{\theta\theta}=\frac{1}{r_i^n}\Bigg\{&\lambda_0 r_k^n\left[\sum_{j=1}^{N}A_{kj}^{(1)}U_j+\frac{U_k}{r_k}\right]+2\mu_0 r_k^{n-1}U_k-l^2\lambda_0 n r_k^{n-1}\left[\sum_{j=1}^{N}A_{kj}^{(2)}U_j+\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(1)}U_j-\frac{U_k}{r_k^2}\right]\\&-l^2\lambda_0 r_k^n\left[\sum_{j=1}^{N}A_{kj}^{(3)}U_j+\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(1)}U_j-\sum_{j=1}^{N}\frac{2}{r_k^2}A_{kj}^{(1)}U_j+\frac{2U_k}{r_k^3}\right]\end{aligned}$$

**Table 1** Material properties of FG micro-cylinder [38]

| $E_0=E_{cer}=151$ (GPa) | $\rho_0=\rho_{cer}=5700\,(kg/m^3)$ |
|---|---|
| $\vartheta=0.3$ | $\alpha_0=18\times10^{-6}\,(1/°C)$ |
| $\lambda_0=20\,GPa$ | $\mu_0=10\,GPa$ |

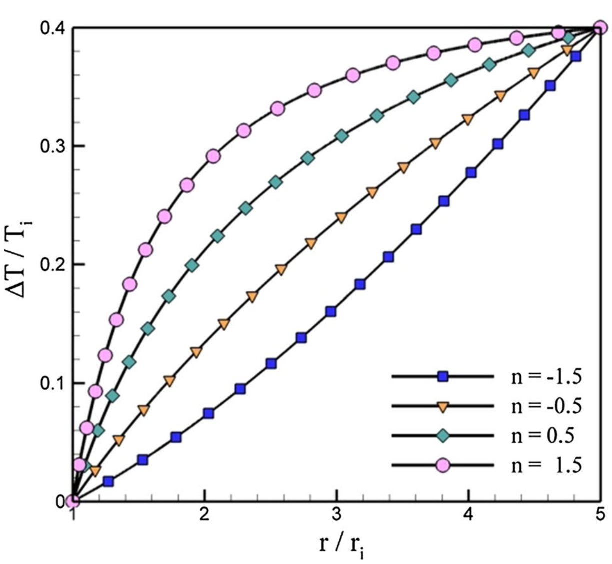


**Fig. 3** Dimensionless temperature distribution in the FG micro-cylinder

$$-2l^2\mu_0 n r_k^{n-1}\left[\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(1)}U_j-\frac{U_k}{r_k^2}\right]$$
$$-2l^2\mu_0 r_k^{n}\left[\sum_{j=1}^{N}\frac{1}{r_k}A_{kj}^{(2)}U_j-\sum_{j=1}^{N}\frac{2}{r_k^2}A_{kj}^{(1)}U_j+\frac{2U_k}{r_k^3}\right]\Bigg\}-\frac{(3\lambda_0+2\mu_0)\,\alpha_0 r_k^{2n}\left(A+Br_k^{-n}\right)}{r_i^{2n}}. \tag{30}$$

Also, using the GDQ method, the radial and hoop strains are written as follows:

$$\varepsilon_{rr}=\sum_{j=1}^{N}A_{kj}^{(1)}U_j, \tag{31}$$

$$\varepsilon_{\theta\theta}=\frac{U_k}{r_k}. \tag{32}$$

Here, the Von Mises stress and strain are defined to evaluate equivalent stresses and strains as:

$$\sigma_{eq}=\sqrt{\sigma_{rr}^2+\sigma_{\theta\theta}^2-\sigma_{rr}\sigma_{\theta\theta}}, \tag{33}$$

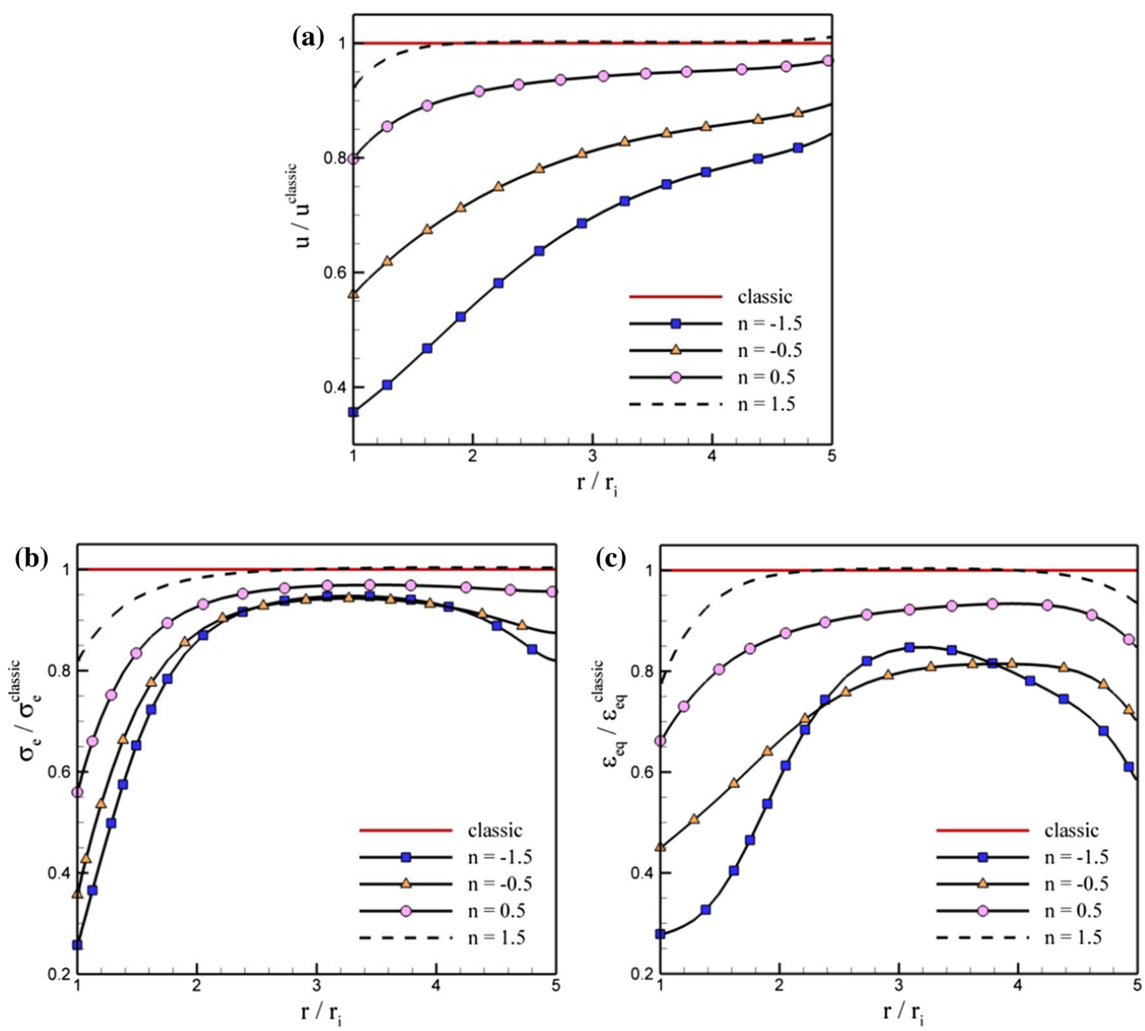


**Fig. 4** Non-homogeneous constant effect on distribution of displacement, stress and strain ($T_i = T_o = 0$, $P_i = P_o = 0$, $\omega = 100\,\text{rad/s}$, $n = 0.5$, $l = 0.4\,\mu\text{m}$). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

$$\varepsilon_{eq} = \sqrt{\varepsilon_{rr}^2 + \varepsilon_{\theta\theta}^2 - \varepsilon_{rr}\varepsilon_{\theta\theta}}. \tag{34}$$

## 6 Numerical results and discussion

In this section, numerical results are presented to study the effects of different parameters on the displacement, stress and strain distributions. At first, it is important to ensure the convergence of the GDQ method for an accurate solution. So, a convergence study is performed in Fig. 1 to determine the non-dimensional displacement, radial and hoop stresses of a rotating micro-cylinder. It is seen that once a sufficient number of nodes ($N$) is used, the displacement and stresses of the micro-cylinder do not change significantly. Therefore, the number of nodes $N = 36$ is used for assurance and increase in accuracy. In order to validate the proposed method, the results are shown for a non-rotating micro-cylinder, regardless of the thermal effect in Fig. 2. For this purpose, inner radius of $r_i = 1\,\mu$m and outer radius of $r_o = 5\,\mu$m are considered. The micro-cylinder are exposed to internal pressure of $P_i = 10\,$MPa and external pressure of $P_\text{o} = 0$. Also, the material length scale parameter and non-homogeneity constant are considered as $l = 0.1\,\mu$m and $n = 0.5$, respectively. Other mechanical and thermal properties of the FG micro-rotating cylinder are listed in Table 1 [38]. It can be seen that the results of the GDQ method have a good agreement with the results presented by Sadeghi et al. [38].

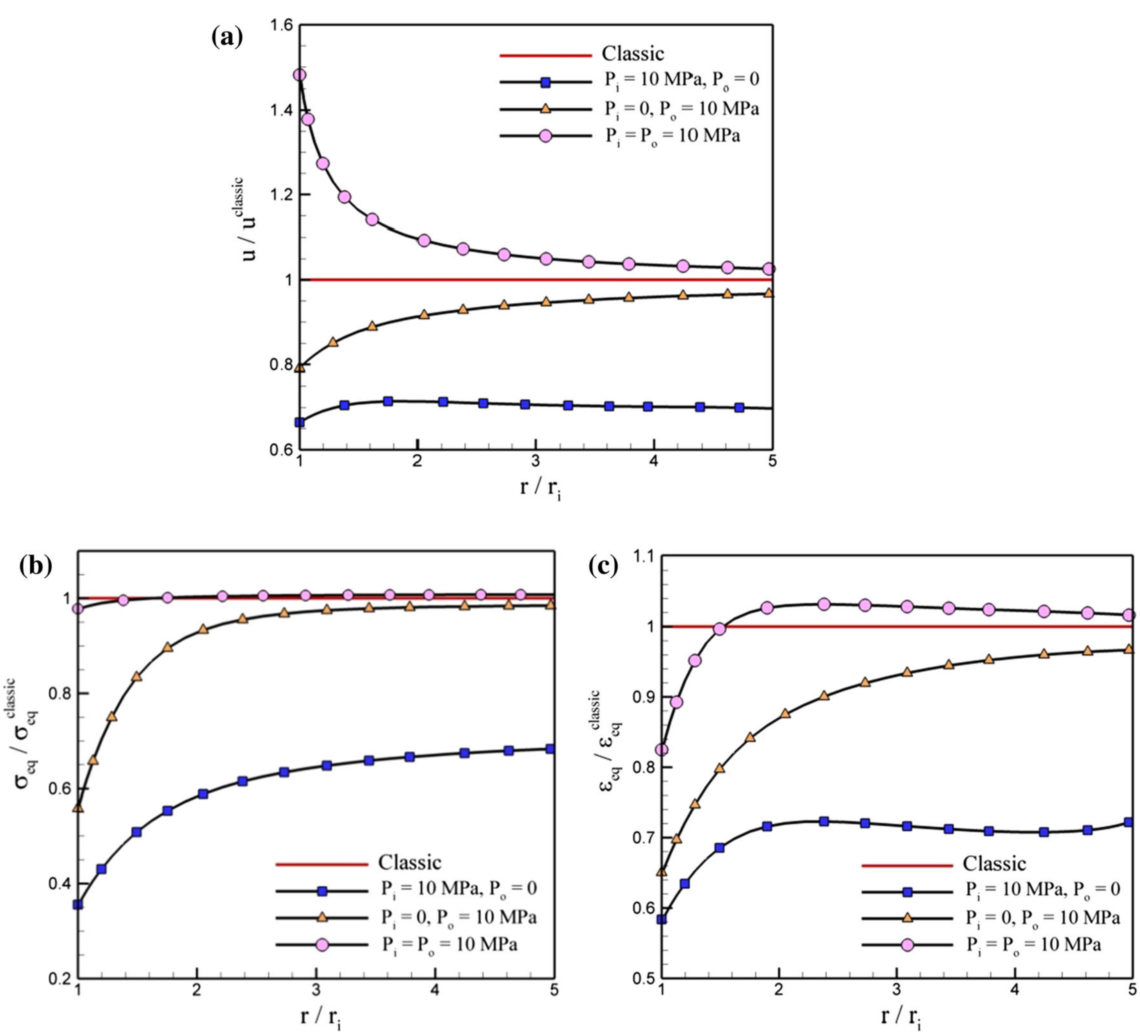


**Fig. 5** Pressure effect on distribution of displacement, stress and strain ($T_i = T_o = 0, \omega = 0, n = 0.5, l = 0.4\,\mu$m). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

The dimensionless temperature distribution for different values of the non-homogeneity constant is shown in Fig. 3. It can be seen that the non-homogeneity constant has significant effects on the temperature distribution in which the temperature increases with increasing non-homogeneity constant.

Dimensionless displacement, equivalent stress and strain distributions in the radial direction of the micro-rotating cylinder with characteristic length parameter of $l = 0.4\,\mu\text{m}$ in the absence of thermal and mechanical loading are depicted in Fig. 4 for different values of the non-homogeneity constant. Each of these three dimensionless parameters is obtained after division of the non-classic parameter by the corresponding classic parameter. As shown in Fig. 4, an increase in the non-homogeneity constant causes the results computed from SGT to increase toward the results of the classical approach ($l = 0$). Therefore, it can be concluded that the strain gradient effect on the displacement, equivalent stress and strain distribution is reduced by an increase in non-homogeneity constant. Also, a stronger effect of strain gradient impact is observed at the inner radius. One of the important findings in this figure is that by changing the non-homogeneity constant, the profile of an FG micro-cylinder can be designed to minimize the stresses in the structure, especially at the inner radius (as shown in Fig. 4b). This provides a powerful tool for designers to make cylinders that can withstand loadings beyond what is currently achievable through natural materials.

In order to assess the effects of the internal and external pressure on the non-classical distributions of displacement, equivalent stress and strain, Fig. 5 is presented. In this figure, three different boundary conditions are investigated. As shown, the effect of the strain gradient on the equivalent stress and strain is more significant for the imposed internal pressure. However, for the micro-cylinder with both external and internal pressure, the

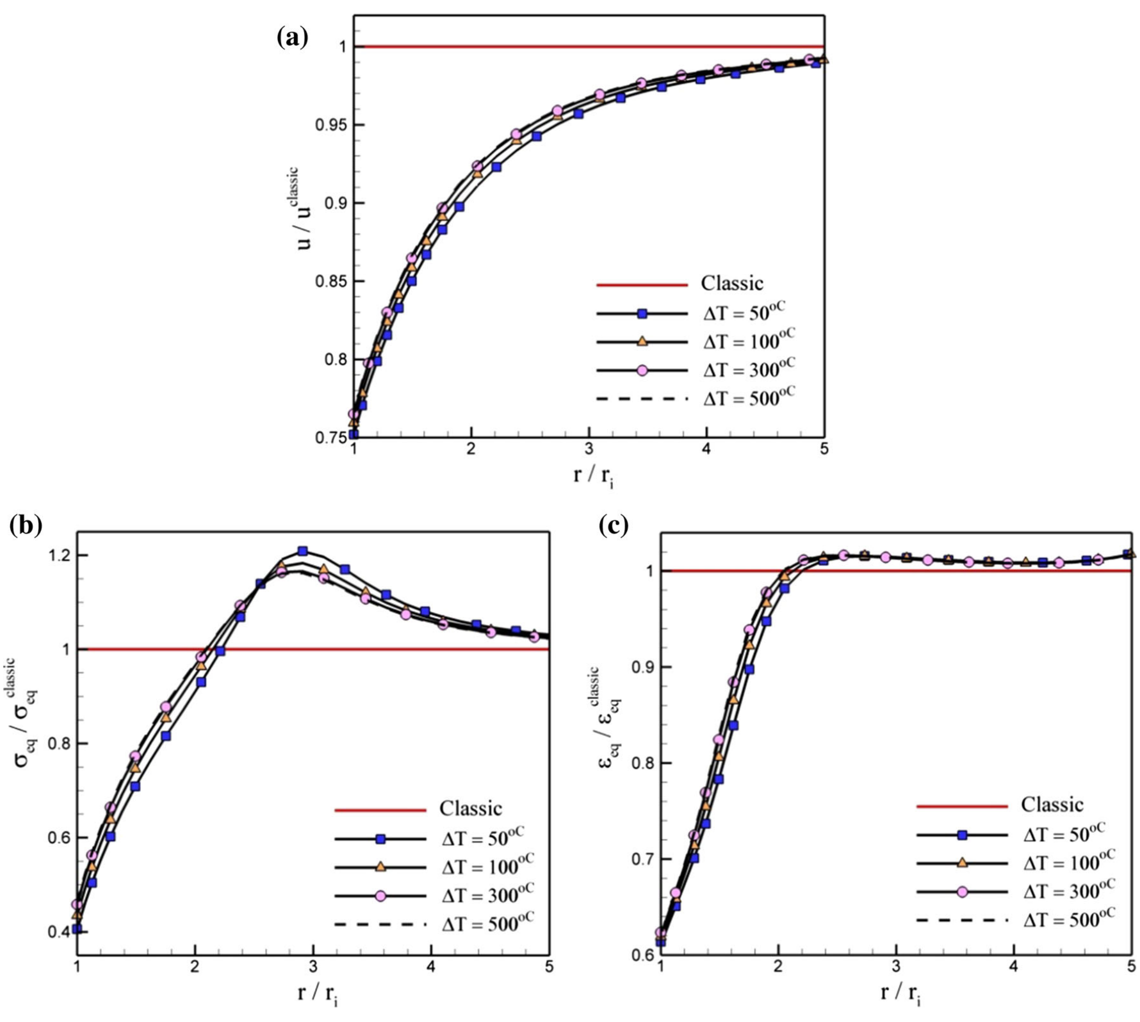


**Fig. 6** Thermal effect on distribution of displacement, stress and strain ($T_i = 25\,^\circ\text{C}$, $P_i = 10\,\text{MPa}$, $P_o = 0$, $\omega = 0$, $n = 0.5$, $l = 0.4\,\mu\text{m}$). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

radial displacement is influenced more significant at the inner radius. If the cylinder is exposed to only internal or external pressure, the strain gradient theory predicts lower displacement, stress and strain compared to the classic theory. However, for a cylinder under both external and internal pressure, the non-classic displacement is predicted to be greater than the classic one and equivalent strain and stress are less influenced by employing strain gradient theory.

The effects of simultaneous temperature field and internal pressure on the distributions of displacement, equivalent stress and strain for a non-homogeneity constant of $n = 0.5$ and characteristic length parameter of $l = 0.4\,\mu\text{m}$ are illustrated in Fig. 6. It is found that an increase in temperature increases displacement, equivalent stress and strain. As shown in Fig. 6, thermal effects reduce the effect of strain gradient impact on the elasticity response of cylinder compared to the cylinder with the only internal pressure. As shown in this figure, over the whole radius range, the strain gradient theory predicts a lower displacement than the classic theory. However, the thermal field causes different behavior to be observed for the strain and the stress response. As shown in Fig. 6b, c, the locations where different curves of stress and strain ratio cross the classical solution are $\frac{r}{r_i} = 2.22281, 2.14968, 2.09913$ and $2.08947$ for $\Delta T = 50, 100, 300$ and $500\,^{\circ}\text{C}$, respectively. So, before and after this cross-points, the stress and strain responses are lower and higher than the corresponding classic response, respectively.

Figure 7 shows the effects of angular velocity in the absence of a thermal field on displacement, equivalent stress and strain for a non-homogeneity constant of $n = 0.5$ and characteristic length parameter of $l = 0.4\,\mu\text{m}$. It can be found that an increase in angular velocity increases the elasticity response ratio of the micro-cylinder.

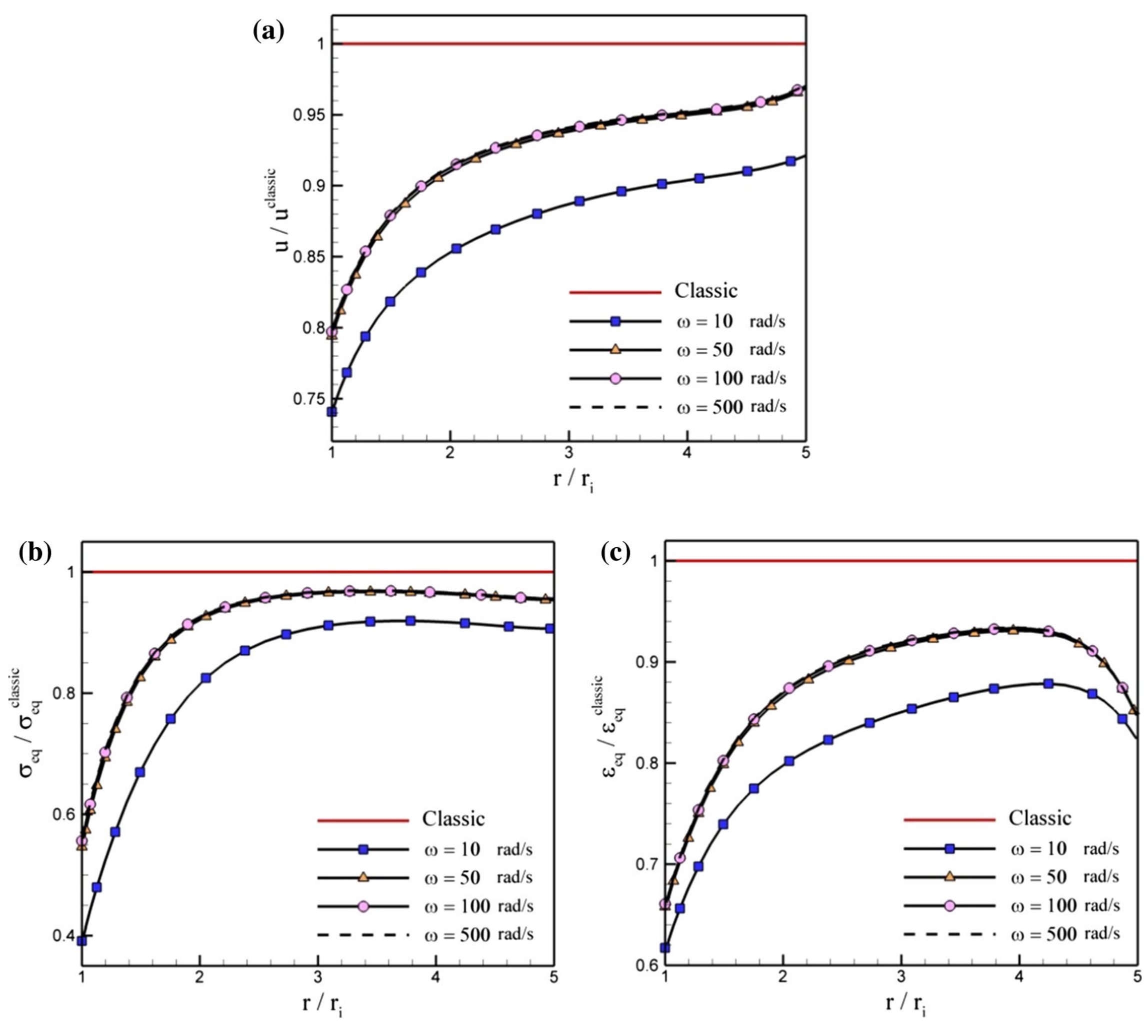


**Fig. 7** Angular velocity effect on distribution of displacement, stress and strain ($T_i = T_o = 0$, $P_i = 10\,\text{MPa}$, $P_o = 0$, $n = 0.5$, $l = 0.4\,\mu\text{m}$). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

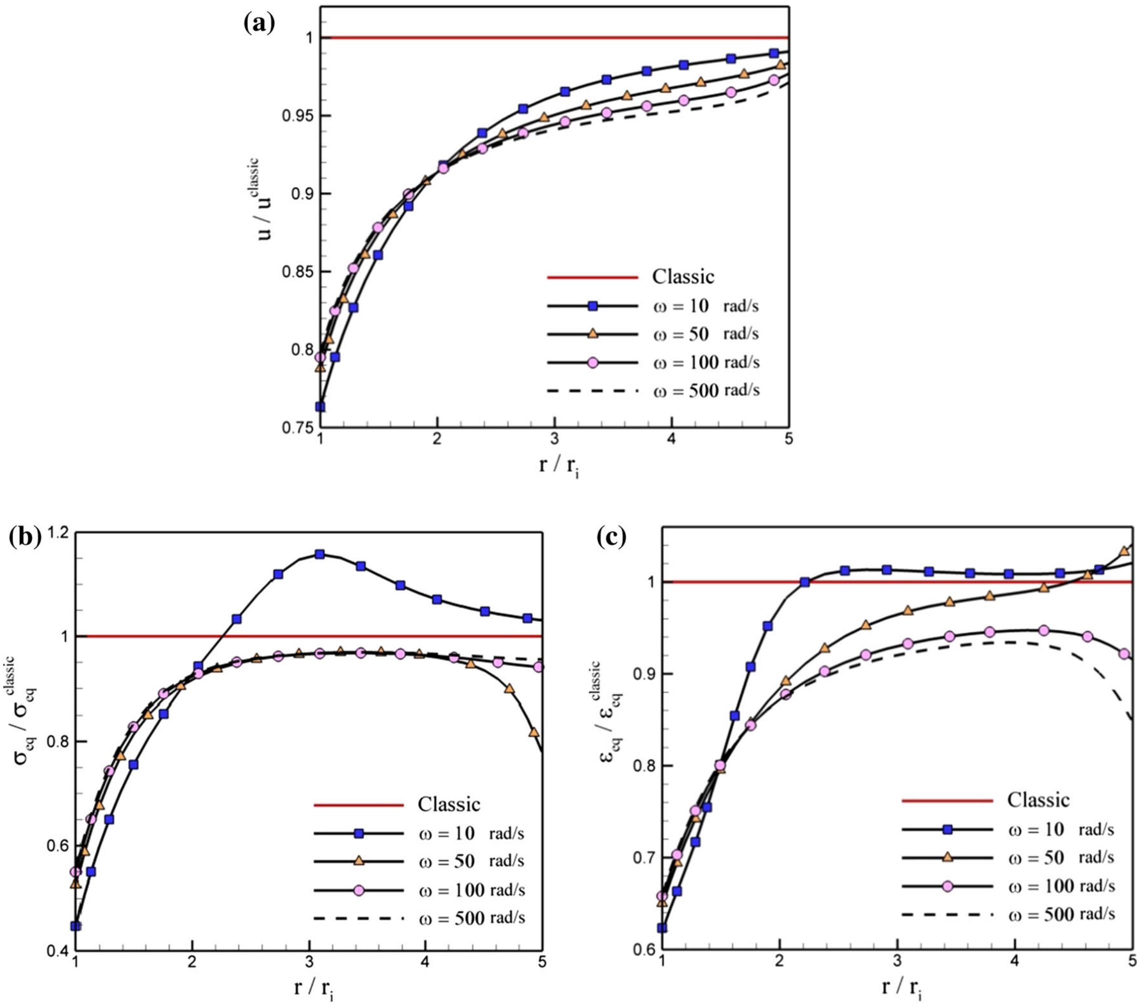


**Fig. 8** Angular velocity effect on distribution of thermal displacement, stress and strain ($T_i = 25\,^\circ$C, $T_o = 125\,^\circ$C, $P_i = 10\,$MPa, $P_o = 0$, $n = 0.5$, $l = 0.4\,\mu$m). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

However, the elasticity response ratio does not change significantly in the high angular velocity range ($\omega > 50\,\mathrm{rad/s}$) in the absence of the thermal effect. This is due to an increasing centrifugal force for higher values of angular velocity which acts to increase the stiffness of the body. The strain gradient effect is more pronounced for a more flexible structure that has high strain gradient conditions under mechanical loading, so increasing the angular velocity tends to reduce this effect.

The effects of angular velocity on displacement, equivalent stress and strain are presented for a micro-rotating cylinder which is placed under thermal and mechanical loading in Fig. 8. In this case, the non-homogeneity constant and characteristic length parameter are considered as $n = 0.5$ and $l = 0.4\,\mu$m, respectively. Considering Fig. 8, the angular velocity $\omega = 10\,\mathrm{rad/s}$ causes effects of the strain gradient on equivalent stress and strain to dominate over the classical state. Moreover, it is clear from a comparison of Figs. 7 and 8 that the thermal effects have a significant impact in the non-classic results.

Size-dependent analysis for a micro-cylinder is performed in Fig. 9. The results of this figure are for a micro-cylinder with constant thickness $4\,\mu$m and various values of the inner radius. It is assumed that $n = 0.5$, $l = 0.4\,\mu$m and $P_i = 10\,$MPa. As depicted in Fig. 9, an increase in the inner radius increases displacement, equivalent stress and strain and reduces the effects of the strain gradient. Also, by increasing the inner radius, the dependency of the results over the whole range of the radius is reduced.

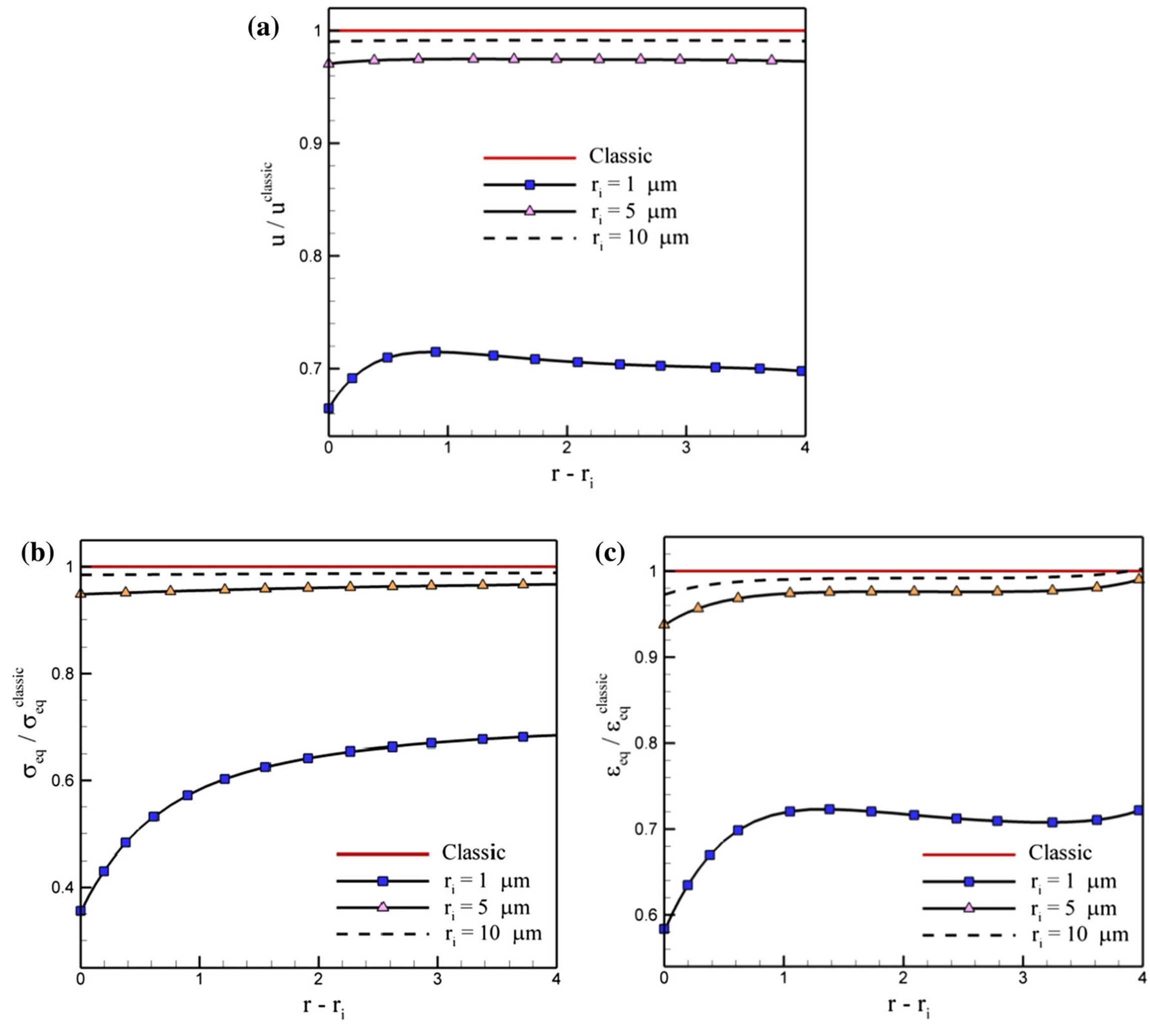


**Fig. 9** Internal radius effect on distribution of displacement, stresses and strain ($T_i = T_o = 0$, $P_i = 10\,\text{MPa}$, $P_o = 0, \omega = 0, r_o = r_i + 4\,\mu\text{m}, n = 0.5, l = 0.4\,\mu\text{m}$). **a** The radial displacement; **b** the Von Mises stress; **c** the Von Mises strain

## 7 Conclusion

In this paper, strain gradient elasticity formulation was expressed for an FG micro-rotating cylinder which is exposed to mechanical and thermal loads. The material properties were assumed to be varying based on a power-law as a function of radial direction. The Fourier heat transfer equation was presented to obtain one-dimensional and axisymmetric temperature distribution in the FG micro-rotating cylinder. The GDQ method was employed to solve the governing thermoelastic equations. Effects of different parameters such as non-homogeneity constant, angular velocity, thermal and mechanical field on distributions of displacement, Von Mises stress and Von Mises strain were studied to compare with classical elasticity results. Regarding the diagrams, the following results can be inferred:

- The non-homogeneity constant has a significant effect on the elastic behavior of micro-rotating cylinder. An increase in dimensionless elastic response includes dimensionless displacement, temperature, equivalent stress caused over the whole range of radius by increasing the non-homogeneity constant at the range of $n > -1.5$. However, this behavior can be observed for dimensionless equivalent strain at $n > -0.5$. Therefore, the elastic response can be optimized in the structure, especially at the inner radius, by proper choice of constituent materials. Also as a consequence, the strain gradient effect is reduced by increasing the non-homogeneity constant.

- It should be pointed out as either internal or external pressure is exposed on the micro-cylinder, the strain gradient elasticity predicts lower displacement, equivalent stress and strain compared to classic theory, whereas when internal pressure and external pressure are simultaneously applied, the effect of strain gradient on the equivalent stress and strain decreases compared to classic theory.
- Considering the diagrams, an increase in temperature increases displacement, equivalent stress and strain. It is found that the thermal field causes the Von Mises stress and strain ratios to become higher than 1 in some range of radius. Therefore, a thermal field has normally improper effects on strain gradient theory.
- In the absence of a thermal field, the elasticity response ratio increases by raising the angular velocity. However, $\omega > 50\,\mathrm{rad/s}$ has not a significant influence on distributions of displacement, equivalent stress and strain ratios. In the presence of thermal field, the strain gradient elasticity predicts higher equivalent stress and strain compared to classic elasticity for angular velocity $\omega = 10\,\mathrm{rad/s}$, whereas in high angular velocity, the strain gradient elasticity predicts a lower elasticity response ratio.
- Size dependence is the most important parameter for analysis of microstructures. In this study, an increase in the inner radius of micro-cylinder leads to the increase in the elasticity response and as a result reduces the strain gradient effects.

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